Covid19 Sentiment Analysis

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1 Introduction

This report shows the some essential information for each step to import, clean, and transform the text data. The analysis also compared predicting power of logistic models and naive Bayes model regarding to Covid tweets.

2 Data

2.1 Data description

In the sample data, "negative" sentiment has the highest frequency among them. The neutral and extreme sentiment are less than the simple positive or negative sentiment.

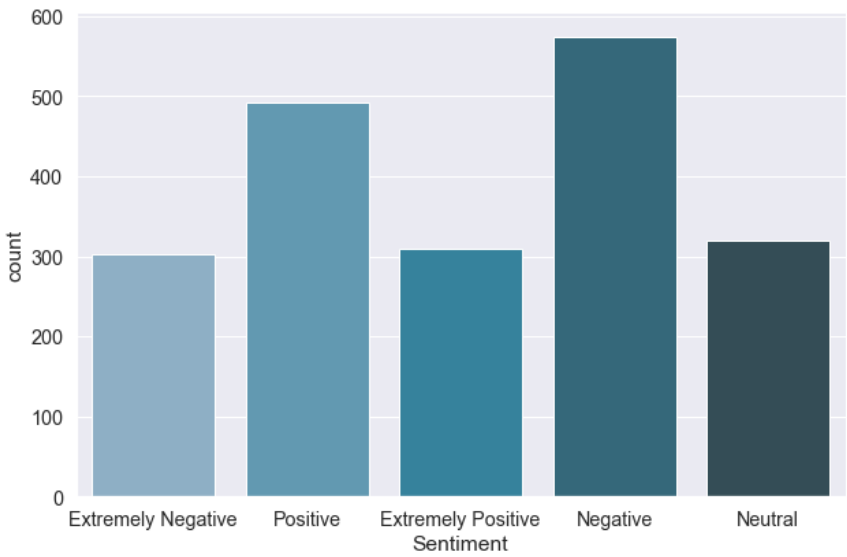

Figure 1: Target variable distribution

2.2 Feature extraction

4 lexicons are used to extract the lexicon-based features. These lexicons includes NRC Word-Emotion Association Lexicon, Opinion Lexicon, AFINN lexicon, and MPQA lexicon. The sentiments are the target for the lexicon based classification models.
Figure 2 provide the distribution for features under AFINN. In "AFINN postive", when the sentiments are 'extremely negative','negetive' and 'neutral', most of the targets fall in and around '0', while the mean and modes of 'extremely positive' and 'positive' are close to 1.2 and 3 respectively. In "AFINN negative", when the sentiments are 'extremely positive','positive' and 'neutral', most of the targets fall in and around '0', while the mean and modes of 'extremely negative' and 'negative' are close to 1.2 and near 4 respectively. Therefore, under AFINN, there are positive relationships to the sentiments. In line with the standard deviations and the figures, the relations are not very strong.

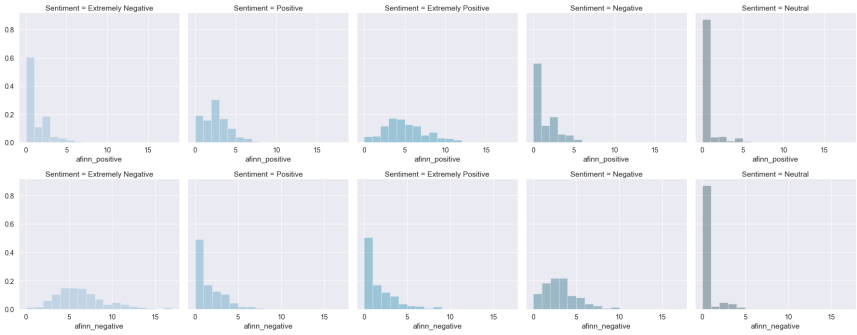

Figure 2: Feature visualization under AFINN

Similarly, with the same logic, there are some feature relation to the sentiments, but the relation are not very strong. See Figure 3.

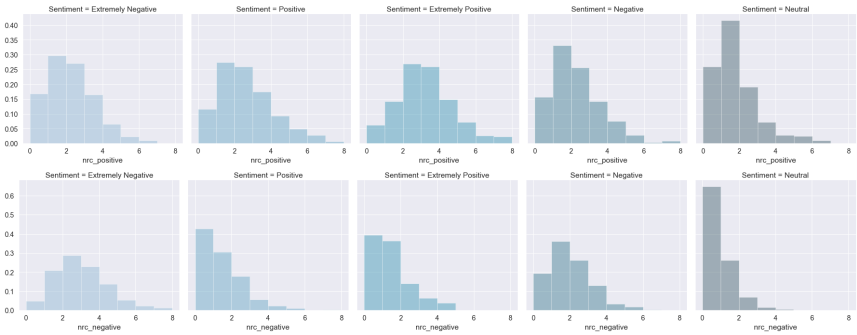

Figure 3: Feature visualization under NRC

Figure 4 illustrated the sentiments by words. Most of the relations between the features and the sentiments are not very strong.

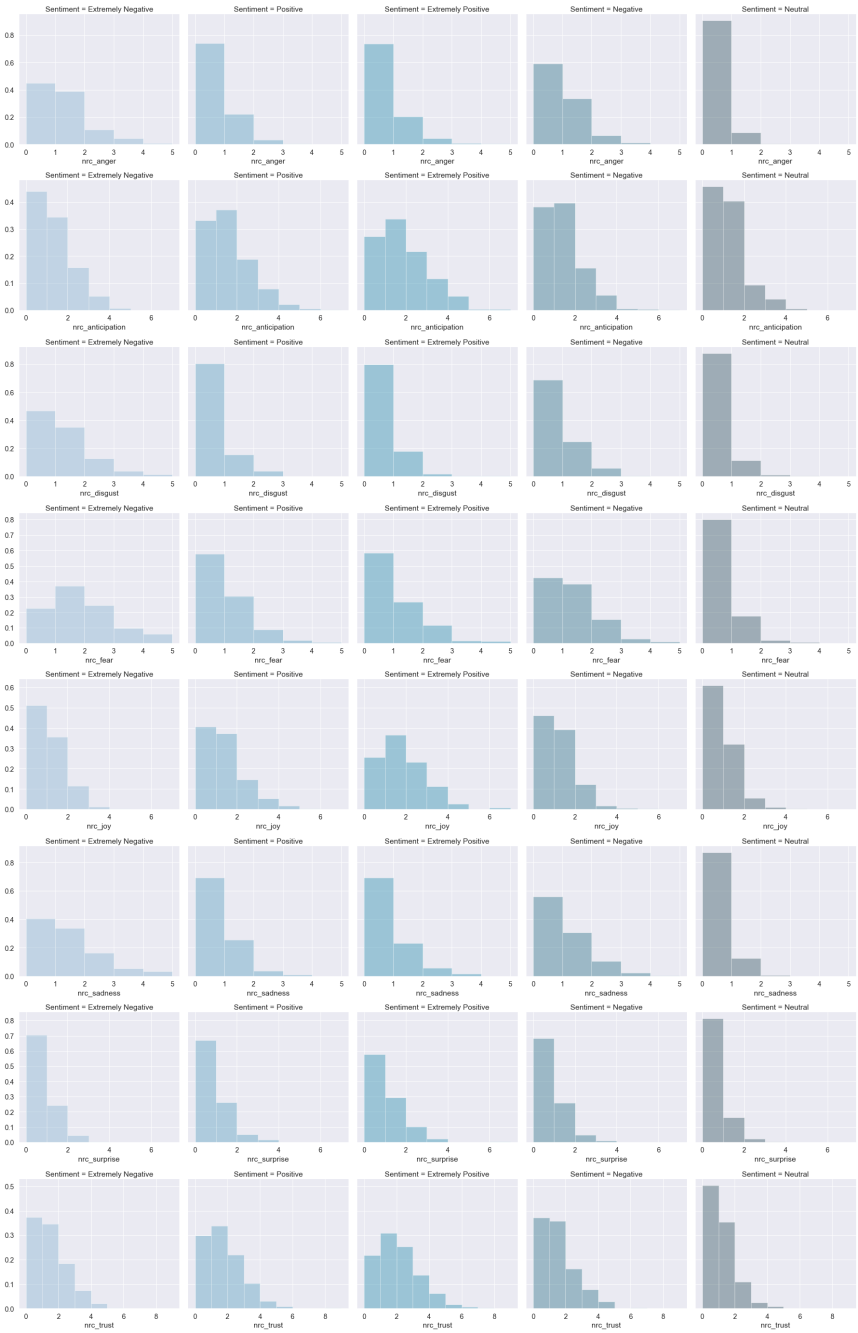

Figure 4: Word-Emotion visualization under NRC

The relation between MPQA lexicon based feature and the sentiments are shown in Figure 5.There are relations between the feature and the sentiments, but the relations are not very strong.

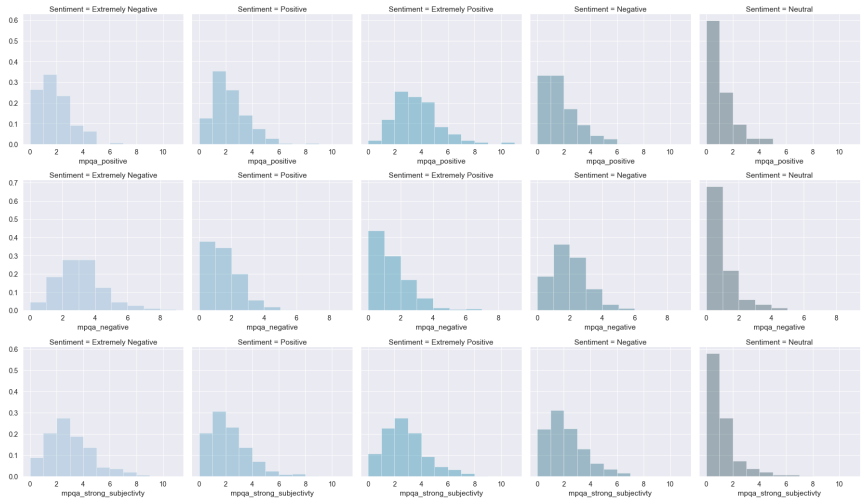

Figure 5: Feature visualization under MPQA

2.3 Data clearance

The clearance has in total 10 steps:

```
Step 1: Remove mentions;
Step 2: Remove urls;
Step 3: Remove punctuation;
Step 4: Remove digits;
Step 5: Remove stop-words;
Step 6: Remove mentions;
Step 7: Replace emojis;
Step 8: Fit and transform data;
Step 9: Apply stemming;
Step 10: Change words into lowercase.
```

From Step 1 to step 6, the clearance remove the information that does not help the analysis. These data are usually useless, but could decrease the operation speed.
The step 7 and step 8 change the format of the data so that the data can be used for further analysis.
The step 9 and step 10 are to treat the difference of a word. Python treats uppercase and lowercase of the same word as different ones, the switch from uppercase to lowercase can change the same words into the same format. Stemming, for example Porter stamming, might not change the word to its original form. Take escape as the example, "escapes" and "escaped" could be both changed into "escap" instead of "escape", leading to a negative influence on the lexicon scoring.

3 Method

3.1 Bag-of-Words (BOW)

The Top 10 words in frequency in presented in Figure 6. The most common word is "covid", of which the frequency is higher than 1600. In BoW, the words are independent to each other. A dictionary can be created based on a certain text and every sentence in the text can be thus represented as a vector based on the frequency of the words in that sentence.Therefore, the result is found reliable after checking the vector created for the word.

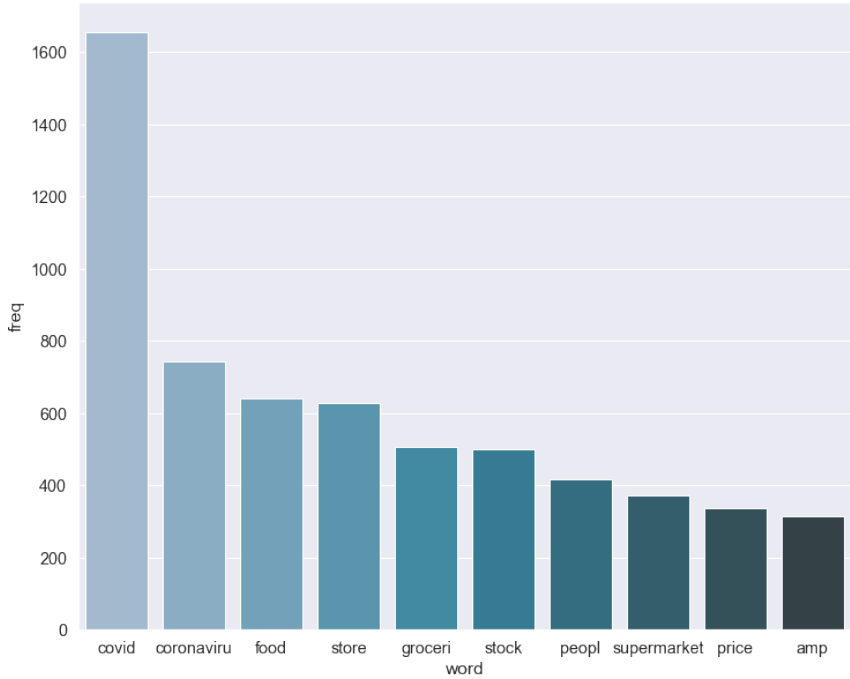

Figure 6: Top 10 words in frequency by BOW

Two other types of representation

Term Frequency Inverse Document Frequency (TFIDF) vectorization

Term Frequency Inverse Document Frequency(TFIDF) is a statistical measure that evaluates the relevance of a word to a document in a bundle of documents by multiplying the times that the words appear and its inverse document frequency across the collection of documents. TFIDF works by proportionally increasing the number of times a word appears in the document but is counterbalanced by the number of documents in which it is present. Hence, words frequently appears across all documents will not rank very high.

However in contract, a word that appears a lot of time in some of the documents set can gain a higher rank in line with its probably indication of the context of the document.
Figure 7 illustrated the Top 10 words in frequency by the TFIDF vectorization. The top 9 are the same words as those by BoW, but in different ranks. Still, the most common word is "covid" with this method. The 10th in the TFIDF is the shop, which differs from the one by BoW.

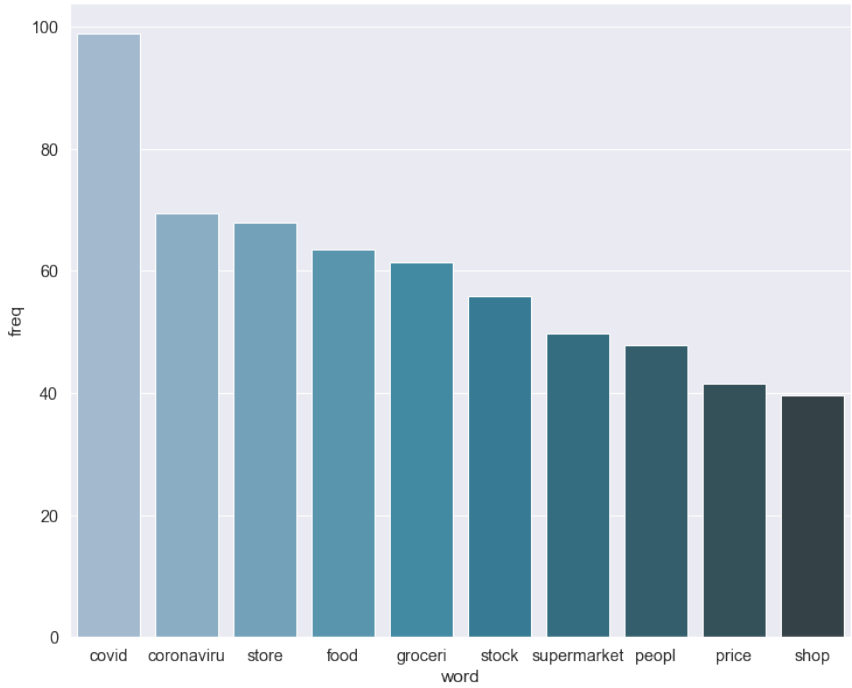

Figure 7: Top 10 words in frequency by TFIDF

The TFIDF has plenty of uses, for instance, for information retrieval, for text summarization and keyword extraction, etc. It is most importantly used in automated text analysis. TFIDF is very useful for scoring words in machine learning algorithms for Natural Language Processing.

3.1.1 Convolutional Neural Networks (CNN)

Convolutional Neural Networks (CNN) is a class of deep learning networks. It is most commonly used to analyze visual imagery. Most of the CNNs are only equivalent. They are applied in image and video recognition, image classification, image segmentation, medical image analysis, natural language processing, brain-computer interfaces and financial time series.
The CNN has three main types of layers: convolutional layer, pooling layer, fully-connected (FC) layer. The convolutional layer is the first layer of the CNN. It is core block of a CNN.

Majority of the computations take place within this layer. The pooling layers conducts dimensionality reduction. It reduces the number of parameters in the input. The FC layer performs classification tasks that based on the features extracted through the previous layers and their different filters.
Figure 8 give an example on how the CNN works.

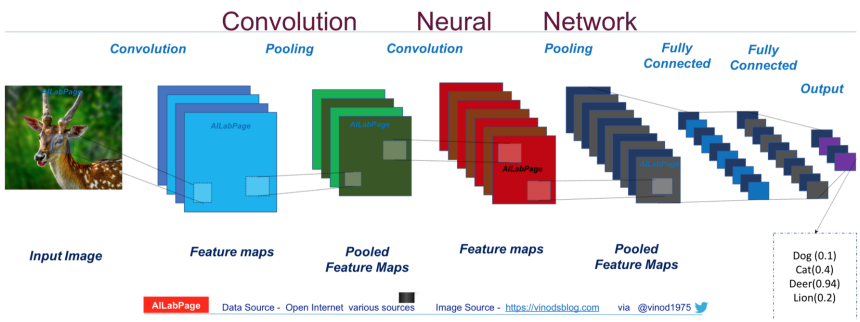

Figure 8: How CNN works

4 Conclusions and discussion

4.1 Results

The Table 1 and Table 2 show the results for logistic regression model and naive Bayes model respectively.

	precision	**recall**	**f1-score**	**support**
Extremely Negative	0.66	0.66	0.66	64
Extremely Positive	0.90	0.56	0.69	66
Negative	0.57	0.65	0.61	110
Neutral	0.68	0.79	0.73	61
Positive	0.55	0.56	0.55	99
accuracy			0.63	400
macro avg	0.67	0.64	0.65	400
weighted avg	0.65	0.63	0.63	400

Table 1: Results to logistic regression

The logistic regression model performs better than the naive Bayes model for each measurement, as a whole and for each sentiment type in this case. This provide the evidence that the logistic regression model has greater predicting ability to the sentiments in this case.

	precision	recall	f1-score	support
Extremely Negative	0.59	0.66	0.62	64
Extremely Positive	0.65	0.50	0.56	66
Negative	0.47	0.53	0.50	110
Neutral	0.68	0.79	0.73	61
Positive	0.76	0.56	0.64	99
accuracy			0.53	400
macro avg	0.57	0.54	0.55	400
weighted avg	0.55	0.53	0.53	400

Table 2: Results to Naive Bayes model

4.2 Features and parameters leading to the results

The features and parameter grid settings for both models are listed in Table 3. The data set are previously separated into training and test subsets. The best parameter grids are identified by "pipeline" and "GridSearchCV" function in python.

Model	Features	Parameter grid settings	
Logistic Regression Model	All the extracted lexicon-based feature represented by BOW	vectorizers (Count and TFIDF) max_df: (Maximum document frequency) 'Features__pipe__vect__max_df': (0.25, 0.5, 0.75) min_df: (How many sentences need to contain the words) 'Features__pipe__vect__min_df': (1,2) Ngram_range: (Both single words and bigrams are used) 'Features__pipe__vect__ngram_range': ((1, 1), (1, 2))	'clf__C': (0.25, 0.5, 1.0) 'clf__penalty': ('l1', 'l2')
Naïve Bayes Model			'clf__alpha': (0.25, 0.5, 0.75)

Table 3: Features and parameters for models

4.3 Usefulness discussion

Both of this the models have accuracy and f1-scores more than 0.5. The models can be used in future analysis for certain purpose, such as evaluating customer's review towards products to predict and make future marketing strategies. However, even the logistic regression model outperforms the naive Bayes model in this case, the values for evaluation criteria are not very high, the model should be improved a little.